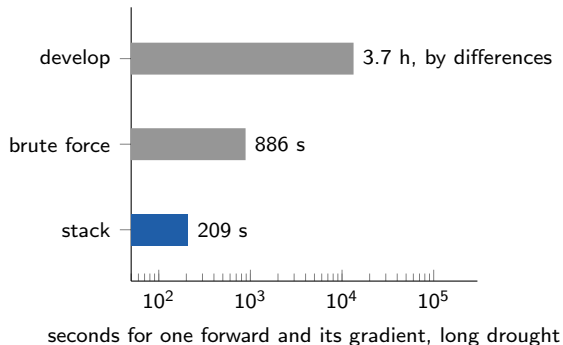


TF24 gradients in plant

What the solver and controller work buys

8 October 2026

A TF24 gradient run costs 0.24–0.43 of brute force



- ▶ **0.24** of brute force at 108 nodes, the pilot included; episodic within 0.40ϵ of it.
- ▶ **0.43** at 215 nodes, which long drought needs to match brute force's accuracy.
- ▶ develop has no gradient: 100 forwards by central differences.
- ▶ Nothing fails across five rainfall records.

Timed alone, one run each. [docs/measurements/headline/](#)

Why: what an evolutionary analysis needs from TF24

- ▶ One singular strategy: **about 84 runs**: resident forwards, invader walks, reverse sweeps.
- ▶ Each answer within ε of converged: 0.025 in $\ln J$, a tenth of each elasticity's spread over eight weather seeds.

Before this work:

- ▶ **No gradients**: about 98 forwards each, by differences.
- ▶ **Walks failed** on episodic and dry rain: 27–38-day steps, pools stable for 26.
- ▶ **Error went unreported**: companions saw 0.33–0.41 of the node error.
- ▶ **Gradients were staircases** at sign changes; curvatures off by up to 13ε .

What changed: four groups, 17 branches, every default off

Gradients	exact counts; the reverse sweep; exact invader replay; the first invasion walks the recording
The split	per-state error weights; a node's step split where its net production changes sign, forward and in the sweep
Settings	<code>control_tf24()</code> : tied tolerance, soil weight, bound, 15-day cap; the soil stepped alone; <code>control_window()</code> from a pilot
Node rule	crowns sorted; the spread (D); walks by birth date

On the aornugent forks, one issue each. The stack sits on odelia 0.5.0 and needs a rebase onto upstream develop (odelia 0.6.2).

The reverse sweep: every gradient for 2.5 forwards

- ▶ One sweep costs **2.5–2.7 forwards** at any node count.
- ▶ Matches central differences to $7e-4$ of the resident's `lma` elasticity.
- ▶ Replaces about 98 forwards of differences per gradient.
- ▶ Invaders replay the resident's recorded steps: at the resident's traits $J' = J$ to the bit.

The split: gradients without staircases

- ▶ Where a cohort's net production changes sign inside a step, its step is cut there.
- ▶ $\ln J$ lies **3–13 $\times$ nearer** the converged answer wherever nodes cross.
- ▶ A $\pm 5\%$ tolerance nudge moves no gradient entry past $\varepsilon/6$; up to 0.52ε without.
- ▶ The sweep through it is within $2e-3$ of central differences on every record.
- ▶ Cost: +5–9% of a gradient run.

Settings: spend accuracy where fitness is earned

- ▶ `control_tf24()` names TF24's setting once; with the window it saves **31–34%** of a gradient run.
- ▶ The **15-day cap** removes the episodic and dry walk failures.
- ▶ The **soil stepped alone** where the stand draws little water: a gradient run takes 0.63 of the time, every elasticity within 0.063ε .
- ▶ The **window**: tolerance loosened once most of J is earned, read off a cheap pilot. On five records nothing fails, $\ln J$ within $5.8e-5$ of a $1e-6$ reference.

The node rule: an error that reports itself

- ▶ **The spread:** each birth-date interval's leaf area spread over point crowns.
- ▶ Uniform nodes converge at the square law: ratios 3.5–4.0 on long-wet and dry.
- ▶ A half-node companion reports **0.87–0.99** of the error, against 0.33–0.41 before.
- ▶ Cost: +6.7% of a gradient run with splits on.
- ▶ Walks by birth date: a shared thinned schedule saves 24–25% of a walk, selection gradient within 0.004ϵ .

Accuracy against brute force, and where nodes matter

against the two-rung answer, long drought	median	largest	time vs brute force
brute force, 215 nodes, 1e-5	0.12ε	0.49ε	1
the stack, 215 nodes	0.13ε	0.53ε	0.43
the stack, 108 nodes	0.50ε	2.1ε	0.24

- ▶ On episodic the stack at 108 nodes is within 0.40ε of brute force on all 50 entries.
- ▶ On long drought 108 nodes are too few; the companion run is what tells you so.

What moves for existing users

- ▶ Every new behaviour is off by default and bit-identical when off.
- ▶ The **height coordinate**, plant's default, does not move.
- ▶ **FF16 on the birth-date coordinate** moves: the deep-crown anchor $17.14 \rightarrow 18.01$. Its error at the default schedule is larger, but falls at the square law, so it can be estimated.
- ▶ TF24 model versions v11–v13.
- ▶ Under default `Control()`, TF24's J on long drought is 0.45 against 12.75 converged: the defaults, not the model, carry the gap.

Open, and how to review

Open

- ▶ Episodic needs nodes finer than its dry spells (past 429).
- ▶ The constant record's later nodes move its gradients up to 3ϵ .
- ▶ Diagnostics next: each run's error estimate, distance from the grid, failures.
- ▶ The radius one grid serves is unmeasured.

Review in four PR groups

1. Gradients (PLANT-93–99)
2. The split (PLANT-100–103, odelia#53–54)
3. Settings (PLANT-104–106, odelia#55)
4. Node rule (PLANT-107–109): the FF16 question